

A Del Pezzo–Blekherman Separation Theorem for Arithmetically Gorenstein Varieties of Quadratic Deficiency One

Corrected and fully detailed note

August 2026

Abstract

Let $X \subset \mathbb{P}_{\mathbb{R}}^{m+c}$ be a geometrically integral, nondegenerate real projective variety with Zariski-dense real locus, and suppose that its homogeneous coordinate ring R is arithmetically Gorenstein with

$$\mathrm{Hilb}_R(t) = \frac{1 + ct + t^2}{(1 - t)^{m+1}}.$$

Thus X has degree $c + 2$ and quadratic deficiency one. We prove the structure of the extreme rays of the dual sums-of-squares cone. Every such ray is either a positive real point evaluation, or its Hankel form has basepoint-free kernel of dimension $m + 1$ and rank c . In the latter case the kernel defines a finite surjective morphism $X \rightarrow \mathbb{P}^m$ of degree $c + 2$. A reduced fiber over a real point has a unique degree-one evaluation relation, all of whose coefficients are nonzero, and degree-two evaluation is an isomorphism after quotienting by the linear equations of the fiber. Positive semidefiniteness forces every such fiber to contain at most one nonreal conjugate pair. We give normalization-independent reciprocal formulas for the evaluation weights in the fully real and one-pair cases, and prove converses producing extreme rays. Consequently every nonnegative non-SOS quadratic section has a basepoint-free extreme separator, and every strictly positive boundary SOS has minimal SOS length exactly $m + 1$. The rank and boundary-length conclusions specialize [2, Proposition 3.5]; the fiberwise formulas extend the mechanism of [1, Theorems 6.1 and 7.1].

1 Set-up and main theorem

Let $X \subset \mathbb{P}_{\mathbb{R}}^{m+c}$ be geometrically integral and nondegenerate, with $m \geq 1$ and $c \geq 1$. Let

$$R = \bigoplus_{d \geq 0} R_d$$

be its standard graded homogeneous coordinate ring, and write $L = \mathcal{O}_X(1)$. We assume throughout that $X(\mathbb{R})$ is Zariski dense and that R is Gorenstein with

$$\mathrm{Hilb}_R(t) = \frac{1 + ct + t^2}{(1 - t)^{m+1}}. \tag{1}$$

In particular, R is Cohen–Macaulay. Scalar extension to $\mathbb{C}$ will be denoted by $R_{\mathbb{C}} = R \otimes_{\mathbb{R}} \mathbb{C}$ and $X_{\mathbb{C}} = X \times_{\mathbb{R}} \mathbb{C}$. Geometric integrality ensures that $R_{\mathbb{C}}$ is a domain. The Cohen–Macaulay and Gorenstein properties are preserved by this field extension; see [5, Chapters 18 and 21].

Define

$$P_X = \{p \in R_2 : p(x) \geq 0 \text{ for every } x \in X(\mathbb{R})\}$$

and

$$\Sigma_X = \left\{ \sum_{j=1}^N q_j^2 : q_j \in R_1, N \geq 1 \right\}.$$

For $\ell \in R_2^*$ let

$$B_\ell(f, g) = \ell(fg), \quad Q_\ell(f) = B_\ell(f, f),$$

and put

$$W_\ell = \text{rad}(B_\ell) = \{f \in R_1 : B_\ell(f, R_1) = 0\}.$$

If $Q_\ell \succeq 0$, then $Q_\ell(f) = 0$ is equivalent to $f \in W_\ell$.

For $x \in X(\mathbb{R})$ choose a nonzero affine lift $\tilde{x}$ and define $\text{ev}_x(h) = h(\tilde{x})$ on R_2 . Replacing $\tilde{x}$ by a nonzero real scalar multiplies ev_x by a positive scalar, so the ray $\mathbb{R}_{\geq 0} \text{ev}_x$ is intrinsic.

Theorem 1.1 (Main theorem). *Under the assumptions above, the following statements hold.*

(i) *The cone Σ_X is closed and*

$$\Sigma_X^* = \{\ell \in R_2^* : Q_\ell \succeq 0\}.$$

(ii) *Every positive real point evaluation spans an extreme ray of Σ_X^* . If ℓ spans any extreme ray of Σ_X^* , then exactly one of the following alternatives occurs:*

- (a) $\ell = \lambda \text{ev}_x$ for some $x \in X(\mathbb{R})$ and $\lambda > 0$;
- (b) W_ℓ has no common zero on $X(\mathbb{C})$ and

$$\dim W_\ell = m + 1, \quad \text{rank } Q_\ell = c.$$

(iii) *In case (ii)(b), the basepoint-free system W_ℓ defines a finite surjective morphism*

$$\varphi_\ell : X \longrightarrow \mathbb{P}(W_\ell^*) \simeq \mathbb{P}^m$$

of degree $c + 2$. There is a nonempty Zariski-open subset of $\mathbb{P}^m$, defined over $\mathbb{R}$ and containing real points, over which the geometric fibers are reduced. Let Γ be a reduced fiber over a real point. Then Γ has $c + 2$ geometric points. Degree-one evaluation has hyperplane image and hence a unique relation up to scalar, every coefficient in this relation is nonzero, and for any m independent real kernel sections $f_1, \dots, f_m$ cutting out the fiber, degree-two evaluation induces an isomorphism

$$(R_{\mathbb{C}})_2 / (f_1, \dots, f_m)(R_{\mathbb{C}})_1 \xrightarrow{\sim} \mathbb{C}^{c+2}.$$

Moreover, Γ contains at most one nonreal conjugate pair.

(iv) *Suppose that $\Gamma = \{p_1, \dots, p_{c+2}\}$ is fully real. After choosing affine lifts, write its unique degree-one relation as*

$$\sum_{i=1}^{c+2} u_i q(p_i) = 0 \quad (q \in R_1),$$

where all u_i are nonzero. If

$$\ell(h) = \sum_{i=1}^{c+2} \mu_i h(p_i),$$

then every μ_i is nonzero, exactly one μ_i is negative, and

$$\sum_{i=1}^{c+2} \frac{u_i^2}{\mu_i} = 0. \quad (2)$$

Conversely, let m real linear sections cut out a reduced fully real finite scheme of length $c+2$, and let u be its unique degree-one relation. Nonzero weights μ_i with exactly one negative entry and satisfying (2) define a rank- c extreme ray with basepoint-free kernel of dimension $m+1$.

(v) Suppose that the reduced fiber consists of real points $p_1, \dots, p_c$ and one pair $z, \bar{z}$. Write

$$\sum_{i=1}^c u_i q(p_i) + v q(z) + \bar{v} q(\bar{z}) = 0$$

with $u_i \in \mathbb{R} \setminus \{0\}$ and $v \in \mathbb{C} \setminus \{0\}$, and write

$$\ell(h) = \sum_{i=1}^c a_i h(p_i) + b h(z) + \bar{b} h(\bar{z}).$$

Then $a_i > 0$, $b \neq 0$, and

$$\sum_{i=1}^c \frac{u_i^2}{a_i} + 2 \operatorname{Re} \left(\frac{v^2}{b} \right) = 0. \quad (3)$$

Conversely, these conditions for a reduced conjugation-stable finite linear section define a rank- c extreme ray with basepoint-free kernel of dimension $m+1$. After rescaling the affine representative of z so that $v = 1$ and writing the correspondingly transformed coefficient as $b = \alpha + i\beta$, the identity becomes

$$\sum_{i=1}^c \frac{u_i^2}{a_i} + \frac{2\alpha}{\alpha^2 + \beta^2} = 0, \quad \alpha < 0.$$

(vi) Every $p \in P_X \setminus \Sigma_X$ is separated from Σ_X by an extreme ray of type (ii)(b). Every strictly positive element of $\partial \Sigma_X$ has minimal SOS length exactly $m+1$.

The proof occupies the remainder of the note. The point-evaluation/basepoint dichotomy is closely related to [2, Lemma 2.3], and the rank and minimal-length conclusions agree with [2, Proposition 3.5]. We include all arguments needed under the stronger Gorenstein Hilbert-series hypothesis.

2 Consequences of the Hilbert series

Proposition 2.1. *One has*

$$\dim R_1 = m + c + 1, \quad \deg X = c + 2,$$

$$\dim(I_X)_2 = \binom{c+1}{2} - 1,$$

and

$$\omega_R \simeq R(1-m), \quad \omega_X \simeq L^{1-m}.$$

In particular, the quadratic deficiency $\binom{c+1}{2} - \dim(I_X)_2$ equals one.

Proof. The coefficient of t in (1) is $m + 1 + c$, and the value of the Hilbert numerator at $t = 1$ is the multiplicity $\deg X = c + 2$. The coefficient of t^2 is

$$\dim R_2 = \binom{m+2}{2} + c(m+1) + 1.$$

Since $X \subset \mathbb{P}^{m+c}$ is nondegenerate,

$$\dim(I_X)_2 = \binom{m+c+2}{2} - \dim R_2 = \binom{c+1}{2} - 1.$$

For a standard graded Gorenstein ring, the a -invariant is the degree of the Hilbert numerator minus the Krull dimension. Here it is $2 - (m+1) = 1 - m$, so $\omega_R \simeq R(1-m)$; sheafification gives $\omega_X \simeq \mathcal{O}_X(1-m) = L^{1-m}$. These canonical-module facts are standard; see [5, Chapter 21]. $\square$

We will use the following reductions repeatedly. The proof spells out the partial-system-of-parameters step that is sometimes left implicit.

Proposition 2.2 (Regular reductions). *Let $R_{\mathbb{C}} = R \otimes_{\mathbb{R}} \mathbb{C}$.*

- (a) *If $f_0, \dots, f_m \in R_1$ have no common zero on $X(\mathbb{C})$, then they form an R -regular sequence and*

$$A = R/(f_0, \dots, f_m)$$

is Artinian Gorenstein with Hilbert series $1 + ct + t^2$. In particular, $\dim A_1 = c$, $\dim A_2 = 1$, and multiplication induces a perfect pairing

$$A_1 \times A_1 \longrightarrow A_2.$$

- (b) *If $f_1, \dots, f_m \in (R_{\mathbb{C}})_1$ have finite projective zero scheme on $X_{\mathbb{C}}$, then they form an $R_{\mathbb{C}}$ -regular sequence and*

$$B = R_{\mathbb{C}}/(f_1, \dots, f_m)$$

is a one-dimensional graded Gorenstein ring with

$$\text{Hilb}_B(t) = \frac{1 + ct + t^2}{1 - t}.$$

Consequently

$$\dim B_0 = 1, \quad \dim B_1 = c + 1, \quad \dim B_d = c + 2 \quad (d \geq 2).$$

Proof. For part (a), the absence of a common projective zero means that the homogeneous ideal $(f_0, \dots, f_m)$ is primary to the irrelevant ideal. Thus the $m + 1$ forms constitute a homogeneous system of parameters of the $(m + 1)$ -dimensional ring R . Since R is Cohen–Macaulay, every homogeneous system of parameters is a regular sequence.

For part (b), let Z be the finite projective zero scheme. Because the ground field is infinite, there is a linear form $f_0 \in (R_{\mathbb{C}})_1$ that does not vanish at any point of the support of Z . Then $f_1, \dots, f_m, f_0$ have no common projective zero and hence form a homogeneous system of parameters of $R_{\mathbb{C}}$. In a Cohen–Macaulay ring every homogeneous system of parameters, in the displayed order, is a regular sequence. Its initial subsequence $f_1, \dots, f_m$ is therefore regular.

A quotient of a Gorenstein ring by a regular sequence is Gorenstein. Multiplying (1) by $(1-t)^{m+1}$ in part (a), and by $(1-t)^m$ in part (b), gives the asserted Hilbert series. Finally, an Artinian graded Gorenstein algebra has one-dimensional socle in its top degree, and multiplication into the socle gives perfect pairings between complementary degrees. In the present Hilbert function $(1, c, 1)$ this is precisely the pairing $A_1 \times A_1 \rightarrow A_2$. The commutative-algebra facts used here are proved in [5, Chapters 17, 18, and 21]. $\square$

3 The dual cone and its kernel faces

Proposition 3.1. *The cone Σ_X is closed and*

$$\Sigma_X^* = \{\ell \in R_2^* : Q_\ell \succeq 0\}.$$

Moreover, Σ_X is full-dimensional in R_2 .

Proof. The dual identity follows from

$$\ell \left(\sum_j q_j^2 \right) = \sum_j Q_\ell(q_j).$$

For closedness, fix a norm on R_1 and let

$$K = \{q^2 : q \in R_1, \|q\| = 1\} \subset R_2.$$

The set K is compact. We claim that $0 \notin \text{conv}(K)$. If $\sum_j \lambda_j q_j^2 = 0$ with $\lambda_j \geq 0$, $\sum_j \lambda_j = 1$, and $\|q_j\| = 1$, then evaluation on $X(\mathbb{R})$ shows that every q_j with $\lambda_j > 0$ vanishes on $X(\mathbb{R})$. Zariski density gives $q_j = 0$ in R_1 , a contradiction. Hence the compact convex set $\text{conv}(K)$ has positive distance δ from the origin. If $r_n k_n$ converges with $r_n \geq 0$ and $k_n \in \text{conv}(K)$, then $r_n \delta \leq \|r_n k_n\|$, so (r_n) is bounded; a convergent subsequence shows that the conical hull of $\text{conv}(K)$ is closed. This conical hull is Σ_X : for a nonzero sum $\sum_j q_j^2$, put $r = \sum_j \|q_j\|^2$ and write it as $r \sum_j (\|q_j\|^2/r)(q_j/\|q_j\|)^2$, omitting zero summands.

The multiplication map $\text{Sym}^2(R_1) \rightarrow R_2$ is surjective because R is standard graded. A surjective linear map is open, so the image of the open cone of positive-definite Gram forms is an open subset of R_2 contained in Σ_X . Thus Σ_X is full-dimensional. $\square$

The following lemma uses the bilinear radical, not merely the zero set of a quadratic polynomial.

Lemma 3.2 (Kernel-face criterion). *Let V be a finite-dimensional real vector space, let $H \subset \text{Sym}^2(V^*)$ be a linear subspace, and put $C = H \cap \text{Sym}_+^2(V^*)$. For $0 \neq Q \in C$, the following are equivalent:*

- (i) Q spans an extreme ray of C ;
- (ii) every $P \in H$ satisfying $\ker Q \subseteq \ker P$ is proportional to Q .

Here kernels are bilinear radicals.

Proof. Suppose first that a nonproportional $P \in H$ satisfies $\ker Q \subseteq \ker P$. Choose a complement V' of $\ker Q$. Both forms vanish whenever one argument belongs to $\ker Q$, while $Q|_{V'}$ is positive definite. Therefore $Q \pm \varepsilon P$ are positive semidefinite for all sufficiently small $\varepsilon > 0$. They give a nontrivial decomposition of the ray of Q , so the ray is not extreme.

Conversely, suppose $Q = Q_1 + Q_2$ with $Q_i \in C$. If $v \in \ker Q$, then $0 = Q(v, v) = Q_1(v, v) + Q_2(v, v)$. Both terms are nonnegative, so each is zero. Cauchy–Schwarz for a positive-semidefinite bilinear form gives $Q_i(v, w) = 0$ for all w , hence $\ker Q \subseteq \ker Q_i$. Condition (ii) then makes each Q_i proportional to Q . $\square$

Because the multiplication map is surjective, the map

$$R_2^* \longrightarrow \text{Sym}^2(R_1^*), \quad \ell \longmapsto B_\ell,$$

is injective. Thus Lemma 3.2 applies directly to Σ_X^* .

Corollary 3.3. *For every $x \in X(\mathbb{R})$, the ray $\mathbb{R}_{\geq 0} \text{ev}_x$ is an extreme ray of Σ_X^* .*

Proof. The form $B_{\text{ev}_x}(f, g) = f(\tilde{x})g(\tilde{x})$ is positive semidefinite of rank one. If a Hankel form P has $\ker B_{\text{ev}_x} \subseteq \ker P$, then P descends to a bilinear form on the one-dimensional quotient $R_1/\ker B_{\text{ev}_x}$ and is therefore proportional to B_{ev_x} . Apply Lemma 3.2 and injectivity of $\ell \mapsto B_\ell$. $\square$

4 Basepoints and the rank theorem

Lemma 4.1 (Basepoints of an extreme kernel). *Let ℓ span an extreme ray of Σ_X^* .*

- (a) *If W_ℓ has a real basepoint $x \in X(\mathbb{R})$, then $\ell = \lambda \text{ev}_x$ for some $\lambda > 0$.*
- (b) *The space W_ℓ has no nonreal basepoint on $X(\mathbb{C})$.*

Proof. For (a), the rank-one form

$$E_x(f, g) = f(\tilde{x})g(\tilde{x})$$

has radical containing W_ℓ . By Lemma 3.2, E_x is proportional to B_ℓ . Positivity and nonzeroness force a positive proportionality constant, and injectivity of $R_2^* \rightarrow \text{Sym}^2(R_1^*)$ gives the claim.

For (b), suppose that a nonreal point $z \in X(\mathbb{C})$ is a common zero of W_ℓ . Choose an affine lift $\tilde{z}$ and define the real Hankel form

$$H_z(f, g) = \text{Re}(f(\tilde{z})g(\tilde{z})).$$

Then $W_\ell \subseteq \ker H_z$. The real evaluation map $R_1 \rightarrow \mathbb{C}$ has real rank two: if its image lay in a real line in $\mathbb{C}$, all homogeneous coordinates of z would have a common complex phase, and z would be a real projective point. Hence there are $f, g \in R_1$ with $f(\tilde{z}) = 1$ and $g(\tilde{z}) = i$. Thus $H_z(f, f) = 1$ and $H_z(g, g) = -1$, so H_z is indefinite. The kernel-face criterion would make it proportional to the positive-semidefinite form B_ℓ , a contradiction. $\square$

We next justify carefully the existence of a full real parameter sequence inside a basepoint-free real linear system.

Lemma 4.2. *Let $W \subseteq R_1$ be a real linear subspace with no common zero on $X(\mathbb{C})$. Then W contains $m + 1$ real sections with no common zero on $X(\mathbb{C})$.*

Proof. The basepoint-free system $W_{\mathbb{C}}$ defines a morphism

$$\psi : X_{\mathbb{C}} \longrightarrow \mathbb{P}(W_{\mathbb{C}}^*)$$

with $\psi^* \mathcal{O}(1) \simeq L_{\mathbb{C}}$. Choose a basis $g_0, \dots, g_s$ of $W_{\mathbb{C}}$, with corresponding target coordinates $T_0, \dots, T_s$. Then

$$\psi^{-1}(D_+(T_i)) = D_+(g_i),$$

and each $D_+(g_i)$ is affine. Thus ψ is affine. The graph factorization through $X_{\mathbb{C}} \times \mathbb{P}(W_{\mathbb{C}}^*)$ shows that ψ is projective, hence proper. To see directly that it is finite, let $V = \text{Spec } A$ be

an affine open in the target and write $\psi^{-1}(V) = \text{Spec } B$. The proper direct-image theorem makes $\psi_* \mathcal{O}_{X_{\mathbb{C}}}|_V$ a coherent $\mathcal{O}_V$ -module [6, Chapter III, Theorem 8.8]; equivalently, B is a finite A -module. Hence ψ is finite. In particular, a finite morphism preserves dimension onto its image, so $\dim \psi(X) = m$.

Set $n = \dim W$ and $k = m + 1$. Consider the incidence variety $\mathcal{I}$ of pairs $(x, U_{\mathbb{C}})$ with $U_{\mathbb{C}} \subseteq \ker(\text{ev}_x)$ in $X_{\mathbb{C}} \times \text{Gr}(k, W_{\mathbb{C}})$. Basepoint-freeness gives $\dim \ker(\text{ev}_x) = n - 1$, and therefore $\dim \mathcal{I} = m + k(n - 1 - k) = k(n - k) - 1 = \dim \text{Gr}(k, W_{\mathbb{C}}) - 1$. Its image in the Grassmannian is closed because the projection is proper, and hence is a proper closed subset. Thus the good subspaces form a nonempty Zariski-open subset, defined over $\mathbb{R}$. The real points of a real Grassmannian are Zariski dense, so this open contains a real point, yielding the required real subspace $U \subset W$. $\square$

Theorem 4.3 (Rank theorem). *If ℓ spans an extreme ray of Σ_X^* and is not a positive real point evaluation, then*

$$\dim W_{\ell} = m + 1, \quad \text{rank } Q_{\ell} = c.$$

Proof. By Lemma 4.1, W_{ℓ} has no common zero on $X(\mathbb{C})$. By Lemma 4.2, choose

$$U = \text{span}_{\mathbb{R}}\{f_0, \dots, f_m\} \subseteq W_{\ell}$$

with no common complex zero. By Proposition 2.2, $A = R/(f_0, \dots, f_m)$ is Artinian Gorenstein with Hilbert series $1 + ct + t^2$.

Because each f_i lies in the bilinear radical of B_{ℓ} , the functional ℓ annihilates UR_1 and induces $\bar{\ell} \in A_2^*$. This induced functional is nonzero, for otherwise ℓ would vanish on all of R_2 . Since A_2 is one-dimensional, the pairing

$$A_1 \times A_1 \longrightarrow \mathbb{R}, \quad (\bar{f}, \bar{g}) \longmapsto \bar{\ell}(\bar{f}\bar{g})$$

is a nonzero scalar multiple of the perfect Gorenstein pairing and is therefore nondegenerate. If $w \in W_{\ell}$, then its class in A_1 pairs to zero with every class in A_1 , so it is zero. Thus $w \in U$, and $W_{\ell} = U$. Finally,

$$\text{rank } Q_{\ell} = \dim R_1 - \dim W_{\ell} = (m + c + 1) - (m + 1) = c.$$

$\square$

5 The kernel morphism and finite linear sections

Fix a non-evaluation extreme ray ℓ and put $W = W_{\ell}$. By Theorem 4.3, W is basepoint-free and $\dim W = m + 1$.

Proposition 5.1. *The system W defines a finite surjective morphism*

$$\varphi : X \longrightarrow \mathbb{P}(W^*) \simeq \mathbb{P}^m$$

with $\varphi^ \mathcal{O}_{\mathbb{P}^m}(1) \simeq L$ and $\deg \varphi = c + 2$. There is a nonempty Zariski-open subset of $\mathbb{P}^m$, defined over $\mathbb{R}$ and containing real points, over which all geometric fibers are reduced.*

Proof. Choose a basis $f_0, \dots, f_m$ of W , with corresponding coordinates $T_0, \dots, T_m$ on the target. For each i ,

$$\varphi^{-1}(D_+(T_i)) = D_+(f_i),$$

and $D_+(f_i)$ is affine by the construction of Proj ; see [6, Chapter II, Section 2]. Thus φ is affine. Its graph factorization through $X \times \mathbb{P}^m$ makes it projective and therefore proper.

As in Lemma 4.2, the proper direct-image theorem [6, Chapter III, Theorem 8.8] then shows on every affine target chart that the coordinate algebra of its inverse image is a finite module. Hence φ is finite.

The image of a finite morphism is closed. Give the scheme-theoretic image its induced integral structure. The resulting map from X onto that image is still finite and surjective; on affine charts its coordinate-ring extension is integral, and integral extensions preserve Krull dimension. Hence the image is a closed irreducible subset of $\mathbb{P}^m$ of dimension m , and must equal $\mathbb{P}^m$. The pullback identity $\varphi^*\mathcal{O}_{\mathbb{P}^m}(1) \simeq L$ is part of the construction from the basepoint-free linear system.

After base change to $\mathbb{C}$, the finite dominant morphism $\varphi_{\mathbb{C}}: X_{\mathbb{C}} \rightarrow \mathbb{P}_{\mathbb{C}}^m$ induces a finite extension of function fields. In characteristic zero this extension is separable. Therefore $\varphi_{\mathbb{C}}$ is étale at the generic point and hence generically étale. Let $E \subset X_{\mathbb{C}}$ be the non-étale locus. It is closed and does not contain the generic point; its image is closed because $\varphi_{\mathbb{C}}$ is finite. Thus $\Omega_{\mathbb{C}} = \mathbb{P}_{\mathbb{C}}^m \setminus \varphi_{\mathbb{C}}(E)$ is a nonempty open over which the whole restriction is finite étale. Every geometric fiber over $\Omega_{\mathbb{C}}$ is therefore a reduced zero-dimensional scheme. The construction of E and $\Omega_{\mathbb{C}}$ is invariant under complex conjugation, so $\Omega_{\mathbb{C}}$ descends to a nonempty open $\Omega \subset \mathbb{P}_{\mathbb{R}}^m$. Finally, $\mathbb{P}^m(\mathbb{R})$ is Zariski dense in $\mathbb{P}_{\mathbb{C}}^m$, and hence $\Omega(\mathbb{R}) \neq \emptyset$.

It remains to compute the degree. Choose a geometric point $y \in \Omega_{\mathbb{C}}$ and make a complex change of basis in $W_{\mathbb{C}}$ so that $y = [1 : 0 : \cdots : 0]$. The pullback of the homogeneous ideal $(T_1, \dots, T_m)$ of y is $(f_1, \dots, f_m)$. Basepoint-freeness ensures that f_0 is nonzero on their common projective zero scheme, so the scheme-theoretic fiber is $\text{Proj } B$, where

$$B = R_{\mathbb{C}}/(f_1, \dots, f_m),$$

which has Hilbert series $(1 + ct + t^2)/(1 - t)$ by Proposition 2.2. Therefore the zero-dimensional scheme $\text{Proj } B$ has length equal to the eventual constant value $c + 2$ of its Hilbert function. On the finite étale open, $\varphi_{\mathbb{C}}$ is finite locally free of constant rank, equal to the function-field degree of φ ; every geometric fiber there has that many reduced points. Consequently $\deg \varphi = c + 2$. $\square$

The next result is stated for an arbitrary finite linear section, so that its converse applications later do not presuppose an already constructed kernel morphism.

Proposition 5.2 (Evaluation on a reduced linear section). *Let $f_1, \dots, f_m \in (R_{\mathbb{C}})_1$ have finite projective zero scheme $\Gamma \subset X_{\mathbb{C}}$, and set*

$$J = (f_1, \dots, f_m), \quad B = R_{\mathbb{C}}/J.$$

Then J is saturated and Γ has scheme-theoretic length $c + 2$. If Γ is reduced, it consists of $N = c + 2$ geometric points and, after choosing affine representatives, the following hold.

- (a) *Degree-one evaluation is injective with hyperplane image:*

$$\text{ev}_1 : B_1 \hookrightarrow \mathbb{C}^N, \quad \dim \text{im}(\text{ev}_1) = N - 1.$$

Thus the evaluations satisfy one relation, unique up to scalar.

- (b) *Every coefficient in this relation is nonzero.*

(c) *Degree-two evaluation is an isomorphism*

$$\text{ev}_2 : B_2 \xrightarrow{\sim} \mathbb{C}^N.$$

Equivalently,

$$(R_{\mathbb{C}})_2/(f_1, \dots, f_m)(R_{\mathbb{C}})_1 \xrightarrow{\sim} \mathbb{C}^N.$$

Proof. By Proposition 2.2, B is one-dimensional Cohen–Macaulay and Gorenstein, with

$$\dim B_1 = c + 1, \quad \dim B_d = c + 2 \quad (d \geq 2).$$

Choose a linear form f_0 that is nonzero on the support of Γ , as in the proof of Proposition 2.2. Its class in the one-dimensional Cohen–Macaulay ring B is a homogeneous parameter and hence a nonzerodivisor. If an element $b \in B$ were annihilated by a power of the irrelevant ideal B_+ , then in particular $f_0^n b = 0$ for some n , forcing $b = 0$. Thus $H_{B_+}^0(B) = 0$. The quotient J^{sat}/J is precisely this irrelevant-torsion submodule of B , so J is saturated. The Hilbert polynomial of the zero-dimensional projective scheme $\text{Proj } B$ is its scheme-theoretic length, and the displayed Hilbert function shows that this constant is $c + 2$; see [6, Chapter I, Section 7].

Assume now that Γ is reduced. By the saturated ideal–subscheme correspondence, J is the full homogeneous ideal of $\text{Proj } B$. For a reduced finite set of projective points this ideal is the intersection of the homogeneous prime ideals of those points; hence J is radical. A homogeneous class evaluates to zero at all chosen affine representatives exactly when its representative belongs to that intersection. Consequently evaluation on the reduced point set is injective in every degree. The dimension statements above give parts (a) and (c).

It remains to prove that the unique degree-one relation has no zero coefficient. Choose a linear form f_0 nonzero at every point of Γ and rescale the affine representatives $p_1, \dots, p_N$ so that $f_0(p_i) = 1$. Let $H = \text{im}(\text{ev}_1) \subset \mathbb{C}^N$. Then H is a hyperplane containing the all-ones vector $\mathbf{1}$, and we may write

$$H = \left\{ x \in \mathbb{C}^N : \sum_{i=1}^N u_i x_i = 0 \right\}.$$

The section f_0 has no zero on $\text{Proj } B$, hence is a homogeneous parameter of the one-dimensional Cohen–Macaulay ring B and therefore a nonzerodivisor. The Artinian reduction $A = B/(f_0)$ is Gorenstein with Hilbert series $1 + ct + t^2$. Under evaluation,

$$A_1 \simeq H/\mathbb{C}\mathbf{1}, \quad A_2 \simeq \mathbb{C}^N/H.$$

Identify $\mathbb{C}^N/H$ with $\mathbb{C}$ by $x \mapsto \sum_i u_i x_i$. Up to a nonzero scalar, multiplication in A gives the pairing

$$([x], [y]) \mapsto \sum_{i=1}^N u_i x_i y_i \quad (x, y \in H).$$

This pairing is perfect. If $u_j = 0$, then $e_j \in H$ and $\sum_i u_i (e_j)_i y_i = 0$ for every $y \in H$. Since $N = c + 2 \geq 3$, the vector e_j is not a scalar multiple of $\mathbf{1}$, so its nonzero class in $H/\mathbb{C}\mathbf{1}$ lies in the radical of the pairing, a contradiction. Thus every u_i is nonzero. $\square$

Remark 5.3. The nonvanishing of the relation coefficients is the Cayley–Bacharach property for this arithmetically Gorenstein set of points. For the general relation between arithmetically Gorenstein point schemes and Cayley–Bacharach phenomena, see [3, 4].

We now specialize again to a reduced real fiber of φ . Such a fiber is stable under conjugation, although its individual points need not be real.

Proposition 5.4 (Evaluation representation). *Let a reduced real fiber be*

$$\Gamma = \{p_1, \dots, p_r, z_1, \bar{z}_1, \dots, z_k, \bar{z}_k\}, \quad r + 2k = c + 2,$$

where the p_i are real. Choose conjugation-compatible affine lifts and write the unique degree-one relation as

$$\sum_{i=1}^r u_i q(p_i) + \sum_{j=1}^k (v_j q(z_j) + \bar{v}_j q(\bar{z}_j)) = 0, \quad (4)$$

with $u_i \in \mathbb{R} \setminus \{0\}$ and $v_j \in \mathbb{C} \setminus \{0\}$. Then there are unique coefficients $\mu_i \in \mathbb{R}$ and $b_j \in \mathbb{C}$ such that

$$\ell(h) = \sum_{i=1}^r \mu_i h(p_i) + \sum_{j=1}^k (b_j h(z_j) + \bar{b}_j h(\bar{z}_j)). \quad (5)$$

All these coefficients are nonzero. More precisely, if $f_0, f_1, \dots, f_m$ is a real basis of W such that the fiber is cut out by $f_1, \dots, f_m$, then for some $\kappa \in \mathbb{R}^\times$,

$$\mu_i = \frac{\kappa u_i}{f_0(p_i)}, \quad b_j = \frac{\kappa v_j}{f_0(z_j)}. \quad (6)$$

Proof. Since $f_1, \dots, f_m \in W_\ell$, the functional ℓ annihilates $(f_1, \dots, f_m)R_1$. The degree-two isomorphism in Proposition 5.2 gives the unique representation (5); reality of ℓ gives conjugate coefficients.

The condition $f_0 \in W_\ell$ says that $\ell(f_0 q) = 0$ for every $q \in R_1$. Substituting (5), the coefficient vector

$$(\mu_i f_0(p_i), b_j f_0(z_j), \bar{b}_j f_0(\bar{z}_j))$$

annihilates the degree-one evaluation hyperplane. Its annihilator is the one-dimensional space spanned by the relation vector in (4). Thus the two vectors differ by a scalar $\kappa \in \mathbb{C}^\times$. Both vectors are fixed by conjugation together with the permutation $z_j \leftrightarrow \bar{z}_j$; conjugating the equality therefore gives $(\kappa - \bar{\kappa})$ times the nonzero relation vector equal to zero. Hence $\kappa \in \mathbb{R}^\times$, proving (6). The relation coefficients and the values of f_0 are nonzero, so all evaluation weights are nonzero. $\square$

6 Inertia and the number of nonreal points

Lemma 6.1 (Hyperplane inertia bound). *Let Q be a nondegenerate real quadratic form on a finite-dimensional real vector space V , and let $H \subset V$ be a hyperplane. If $Q|_H$ is positive semidefinite, then the negative index of inertia of Q is at most one.*

Proof. If $N \subset V$ were a negative-definite subspace of dimension at least two, then $\dim(N \cap H) \geq \dim N - 1 \geq 1$. Any nonzero vector in $N \cap H$ would have negative Q -value, contradicting positive semidefiniteness on H . $\square$

Proposition 6.2. *Every reduced fiber of φ over a real point contains at most one nonreal conjugate pair.*

Proof. Use the notation of Proposition 5.4. On the real vector space $V = \mathbb{R}^r \oplus \mathbb{C}^k$ consider

$$\tilde{Q}(x, w) = \sum_{i=1}^r \mu_i x_i^2 + 2 \operatorname{Re} \left(\sum_{j=1}^k b_j w_j^2 \right).$$

The real degree-one evaluation image is a hyperplane $H \subset V$, and Q_ℓ is the pullback of $\tilde{Q}|_H$. Hence $\tilde{Q}|_H$ is positive semidefinite. All coefficients are nonzero by Proposition 5.4, so $\tilde{Q}$ is nondegenerate. For each j , the form $w \mapsto 2 \operatorname{Re}(b_j w^2)$ on $\mathbb{C} \simeq \mathbb{R}^2$ has determinant $-4|b_j|^2$ in a suitable real basis and hence signature $(1, 1)$. Thus the negative index of $\tilde{Q}$ is at least k . By Lemma 6.1, $k \leq 1$. $\square$

This is the one-relation instance of the inertia mechanism used in [1, Corollary 4.3].

7 Coefficient classifications and converses

Lemma 7.1 (Diagonal form on a hyperplane). *Let $u_i, \mu_i \in \mathbb{R} \setminus \{0\}$, let*

$$H = \left\{ x \in \mathbb{R}^N : \sum_{i=1}^N u_i x_i = 0 \right\}, \quad Q(x) = \sum_{i=1}^N \mu_i x_i^2.$$

Then $Q|_H$ is positive semidefinite with one-dimensional radical if and only if exactly one μ_i is negative and

$$\sum_{i=1}^N \frac{u_i^2}{\mu_i} = 0.$$

In that case

$$\operatorname{rad}(Q|_H) = \mathbb{R} \left(\frac{u_1}{\mu_1}, \dots, \frac{u_N}{\mu_N} \right).$$

Proof. If $Q|_H$ is positive semidefinite, Lemma 6.1 shows that Q has at most one negative coefficient. It cannot have none, because all coefficients are nonzero and then $Q|_H$ would be positive definite. Thus exactly one is negative. Since Q is nondegenerate, the Q -orthogonal complement of H is spanned by

$$\rho = \left(\frac{u_1}{\mu_1}, \dots, \frac{u_N}{\mu_N} \right).$$

The restriction has nonzero radical precisely when $\rho \in H$, which is the reciprocal identity; the radical is then $\mathbb{R}\rho$.

Conversely, relabel so that $\mu_N < 0$ and $\mu_i > 0$ for $i < N$. Eliminating x_N by the equation of H and using the reciprocal identity gives

$$Q|_H = \sum_{i < N} \mu_i x_i^2 - \frac{(\sum_{i < N} u_i x_i)^2}{\sum_{i < N} u_i^2 / \mu_i}.$$

Weighted Cauchy–Schwarz makes this nonnegative, with equality exactly when x_i is proportional to u_i / μ_i . $\square$

Theorem 7.2 (Fully real fibers). *Let $\Gamma = \{p_1, \dots, p_{c+2}\}$ be a fully real reduced fiber and write*

$$\sum_i u_i q(p_i) = 0, \quad \ell(h) = \sum_i \mu_i h(p_i).$$

Then every u_i and μ_i is nonzero, exactly one μ_i is negative, and

$$\sum_i \frac{u_i^2}{\mu_i} = 0.$$

Conversely, let $f_1, \dots, f_m \in R_1$ cut out a reduced fully real scheme Γ of length $c+2$, let u be its unique degree-one relation, and let nonzero coefficients μ_i have exactly one negative entry and satisfy the reciprocal identity. Then

$$\ell(h) = \sum_i \mu_i h(p_i)$$

spans an extreme ray of Σ_X^* , has rank c , and has basepoint-free kernel of dimension $m+1$.

Proof. Choose a real basis $f_0, f_1, \dots, f_m$ of W adapted to the fiber and put

$$U = \text{span}\{f_1, \dots, f_m\}.$$

The degree-one evaluation isomorphism in Proposition 5.2 is conjugation-equivariant. Taking fixed points identifies the real quotient R_1/U with

$$H = \{x \in \mathbb{R}^{c+2} : u \cdot x = 0\}.$$

The form Q_ℓ descends to $\sum_i \mu_i x_i^2$ on H . Its radical on H is one-dimensional because $\dim W_\ell - \dim U = (m+1) - m = 1$. The first assertion follows from Lemma 7.1 and Proposition 5.4.

For the converse, the same conjugation-equivariant evaluation isomorphism identifies R_1/U with the real hyperplane H . Lemma 7.1 then gives a positive-semidefinite form on H with radical generated by

$$\rho = \left(\frac{u_i}{\mu_i} \right)_i \in H.$$

Since $R_1/U \xrightarrow{\sim} H$, the vector ρ is the evaluation vector of some real $f_0 \in R_1$. Every coordinate of ρ is nonzero, so $W = U + \mathbb{R}f_0$ has no common complex zero: every common zero of U belongs to Γ , while f_0 is nonzero at every point of Γ . The pullback Q_ℓ is positive semidefinite and has kernel W , hence rank c .

Finally, W is a homogeneous system of parameters, so Proposition 2.2 gives

$$\dim R_2/W R_1 = 1.$$

The vector space of Hankel forms whose bilinear radical contains W is the annihilator of $W R_1$ in R_2^* and is therefore one-dimensional. The kernel-face criterion, Lemma 3.2, proves extremality. $\square$

Lemma 7.3 (A Lorentzian hyperplane). *Let $V = \mathbb{R}^r \oplus \mathbb{C}$ and*

$$Q(x, w) = \sum_{i=1}^r a_i x_i^2 + 2 \operatorname{Re}(bw^2), \quad a_i > 0, \quad b \in \mathbb{C}^\times.$$

Let

$$L(x, w) = \sum_{i=1}^r u_i x_i + 2 \operatorname{Re}(vw)$$

be a nonzero real linear functional and put $H = \ker L$. Then Q has signature $(r + 1, 1)$, and $Q|_H$ is positive semidefinite with one-dimensional radical if and only if

$$\sum_{i=1}^r \frac{u_i^2}{a_i} + 2 \operatorname{Re} \left(\frac{v^2}{b} \right) = 0. \quad (7)$$

In that case the radical is generated by the nonzero vector

$$\rho = \left(\frac{u_1}{a_1}, \dots, \frac{u_r}{a_r}, \frac{v}{b} \right).$$

Proof. The complex block $2 \operatorname{Re}(bw^2)$ has signature $(1, 1)$, so Q has signature $(r + 1, 1)$. Its associated bilinear form satisfies

$$B_Q(\rho, (x, w)) = L(x, w).$$

Because $L \neq 0$, the vector ρ is nonzero. Moreover,

$$Q(\rho) = L(\rho) = \sum_i \frac{u_i^2}{a_i} + 2 \operatorname{Re} \left(\frac{v^2}{b} \right).$$

Thus (7) is equivalent to ρ being a nonzero null vector. If $Q|_H$ has nonzero radical, then

$$0 \neq \operatorname{rad}(Q|_H) = H \cap H^{\perp_Q} = H \cap \mathbb{R}\rho,$$

so $\rho \in H$ and $Q(\rho) = 0$. Conversely, in a quadratic space of signature $(r + 1, 1)$, the orthogonal complement of a nonzero null vector is positive semidefinite with radical equal to the null line. Indeed, in coordinates $Q = t_0^2 - t_1^2 + t_2^2 + \dots + t_{r+1}^2$ one may take the null vector to be $e_0 + e_1$; its orthogonal complement consists of $s(e_0 + e_1) + \sum_{j \geq 2} t_j e_j$, on which $Q = \sum_{j \geq 2} t_j^2$. $\square$

Theorem 7.4 (One conjugate pair). *Let a reduced fiber be*

$$\Gamma = \{p_1, \dots, p_c, z, \bar{z}\}.$$

Write

$$\sum_{i=1}^c u_i q(p_i) + v q(z) + \bar{v} q(\bar{z}) = 0,$$

$$\ell(h) = \sum_{i=1}^c a_i h(p_i) + b h(z) + \bar{b} h(\bar{z}).$$

Then

$$a_i > 0, \quad b \neq 0,$$

and

$$\sum_{i=1}^c \frac{u_i^2}{a_i} + 2 \operatorname{Re} \left(\frac{v^2}{b} \right) = 0.$$

Conversely, let m real linear sections cut out a reduced conjugation-stable scheme consisting of c real points and one conjugate pair. If the coefficients satisfy the displayed conditions, then they define a rank- c extreme ray with basepoint-free kernel of dimension $m + 1$.

After replacing the affine lift of z by v times that lift, the relation coefficient at z becomes one. If the transformed coefficient is $b' = \alpha + i\beta$, then

$$\sum_{i=1}^c \frac{u_i^2}{a_i} + \frac{2\alpha}{\alpha^2 + \beta^2} = 0, \quad \alpha < 0.$$

Proof. Let U be the span of the m linear sections cutting out the fiber. The complex degree-one evaluation isomorphism is conjugation-equivariant; its fixed part identifies R_1/U with the real hyperplane

$$H = \left\{ (x, w) \in \mathbb{R}^c \oplus \mathbb{C} : \sum_i u_i x_i + 2 \operatorname{Re}(vw) = 0 \right\}.$$

The descended quadratic form is

$$Q(x, w) = \sum_i a_i x_i^2 + 2 \operatorname{Re}(bw^2).$$

Every coefficient is nonzero by Proposition 5.4. The complex block contributes one negative direction. Since $Q|_H$ is positive semidefinite, Lemma 6.1 shows that no a_i can be negative; hence all $a_i > 0$. The restriction has one-dimensional radical because $\dim W_\ell - \dim U = 1$. The relation functional defining H is nonzero (in fact all of its coefficients are nonzero), so Lemma 7.3 gives the reciprocal identity.

Conversely, Lemma 7.3 gives a positive-semidefinite restriction with radical vector

$$\rho = \left(\frac{u_i}{a_i}, \frac{v}{b} \right) \in H.$$

This conjugation-compatible vector is the evaluation vector of a real section f_0 . All its coordinates are nonzero, so $U + \mathbb{R}f_0$ is basepoint-free. The rank and extremality arguments are identical to those in Theorem 7.2: the kernel has dimension $m + 1$, the rank is c , and the space of Hankel forms containing that kernel in their radical is one-dimensional.

For the normalization, multiplying the affine lift of z by v changes the relation coefficient to one and changes the quadratic evaluation coefficient to $b' = b/v^2$. Hence

$$2 \operatorname{Re} \left(\frac{v^2}{b} \right) = 2 \operatorname{Re} \left(\frac{1}{b'} \right) = \frac{2\alpha}{\alpha^2 + \beta^2}.$$

The sum of the real terms is strictly positive, so the last summand is negative, which forces $\alpha < 0$. $\square$

8 Separation and strictly positive boundary sums of squares

Lemma 8.1 (Extreme supporting functionals). *Let $C \subsetneq V$ be a closed full-dimensional convex cone and let $\rho \in \operatorname{int}(C)$. Then*

$$K = \{\lambda \in C^* : \lambda(\rho) = 1\}$$

is compact. A ray $\mathbb{R}_{\geq 0}\lambda$ is extreme in C^ if and only if λ is an extreme point of K . If $v \notin C$, there is an extreme point $\lambda \in K$ with $\lambda(v) < 0$. If $v \in \partial C$, there is an extreme point $\lambda \in K$ with $\lambda(v) = 0$.*

Proof. Choose $\varepsilon > 0$ such that $\rho + \varepsilon B \subset C$, where B is the unit ball. Every nonzero $\lambda \in C^*$ is strictly positive on ρ : if $\lambda(\rho) = 0$, then applying λ to $\rho \pm \varepsilon u \in C$ for all unit u gives $\lambda(u) = 0$, hence $\lambda = 0$. If $\lambda \in K$, then

$$0 \leq \lambda(\rho \pm \varepsilon u) = 1 \pm \varepsilon \lambda(u),$$

so $|\lambda(u)| \leq \varepsilon^{-1}$. Thus K is bounded; it is closed and therefore compact.

Normalizing a conical decomposition by evaluation at ρ shows directly that extreme points of K correspond to extreme rays of C^* . If $v \notin C$, strict separation supplies a nonzero $\lambda \in C^*$ with $\lambda(v) < 0$; after normalization, the minimum of $\mu \mapsto \mu(v)$ on K is negative and is attained on a nonempty face. An extreme point of that face is an extreme point of K . The boundary case is the same, beginning with a supporting functional and the face $\{\lambda \in K : \lambda(v) = 0\}$. $\square$

Theorem 8.2 (Complete separation). *Every $p \in P_X \setminus \Sigma_X$ is separated from Σ_X by a basepoint-free extreme ray $\ell \in \Sigma_X^*$ satisfying*

$$\ell(p) < 0, \quad \dim W_\ell = m + 1, \quad \text{rank } Q_\ell = c.$$

Proof. The cone Σ_X is closed and full-dimensional by Proposition 3.1. Apply Lemma 8.1 to obtain an extreme separator ℓ . It cannot be a positive point evaluation because $p \in P_X$. The conclusion follows from Lemma 4.1 and Theorem 4.3. $\square$

For $p \in \Sigma_X$, its SOS length is the least number of nonzero squares in an SOS representation. Equivalently, it is the minimum rank of a positive-semidefinite Gram form representing p .

Theorem 8.3 (Strictly positive boundary SOS). *If $p \in \partial\Sigma_X$ and $p(x) > 0$ for all $x \in X(\mathbb{R})$, then its minimal SOS length is exactly $m + 1$.*

Proof. By Lemma 8.1, choose an extreme supporting ray $\ell \in \Sigma_X^*$ with $\ell(p) = 0$. Since p is strictly positive, ℓ is not a point evaluation. Hence $W = W_\ell$ is basepoint-free of dimension $m + 1$.

For any representation $p = \sum_j q_j^2$,

$$0 = \ell(p) = \sum_j Q_\ell(q_j),$$

and every summand is nonnegative. Thus every $q_j \in W$. A Gram form on the $(m + 1)$ -dimensional space W has rank at most $m + 1$, so spectral decomposition gives a representation of p by at most $m + 1$ squares.

Suppose, for contradiction, that the minimal length is $r \leq m$, and choose a minimal representation

$$p = q_1^2 + \cdots + q_r^2.$$

The q_i are linearly independent, for otherwise the associated Gram form could be diagonalized with smaller rank. They have no common real zero because p is strictly positive. Extend their span to an m -dimensional subspace $U \subset W$.

A hyperplane $U \subset W$ is the kernel of a unique point $y_0 \in \mathbb{P}(W^*)$. The common zero scheme of U is exactly the fiber $\varphi^{-1}(y_0)$: at a point x , all sections in U vanish precisely when the evaluation functional $W \rightarrow L_x$ is proportional to the quotient functional $W \rightarrow W/U$. The fiber has no real point because it is contained in the common zero set of the q_i . Hence

$$y_0 \notin \varphi(X(\mathbb{R})).$$

The set $\varphi(X(\mathbb{R}))$ is compact, so its complement contains a Euclidean open neighborhood N of y_0 .

By Proposition 5.1, reduced fibers occur over the real points of a nonempty Zariski-open subset $\Omega \subset \mathbb{P}^m$ defined over $\mathbb{R}$. The complement of $\Omega(\mathbb{R})$ is a proper real algebraic subset and has empty Euclidean interior: a real polynomial vanishing on a Euclidean open set is

zero. Thus $N \cap \Omega(\mathbb{R}) \neq \emptyset$. Choose y in this intersection. The reduced real fiber $\varphi^{-1}(y)$ has no real point and consists of $c + 2 \geq 3$ geometric points.

A conjugation-stable reduced finite set of at least three points with no real point contains at least two conjugate pairs: an odd cardinality would force a real point, while an even cardinality at least four gives at least two pairs. This contradicts Proposition 6.2. Therefore the SOS length is not at most m , and hence equals $m + 1$. $\square$

The rank and length results in Theorems 4.3 and 8.3 coincide with the corresponding conclusions of [2, Proposition 3.5]. The argument above contains explicitly the perturbation to a reduced real fiber.

9 Examples and scope

Example 9.1 (Classical Veronese cases). For $(X, L) = (\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(3))$,

$$\sum_{d \geq 0} \binom{3d+2}{2} t^d = \frac{1+7t+t^2}{(1-t)^3}.$$

Thus $(m, c) = (2, 7)$, a non-evaluation extreme ray has rank seven, and a reduced kernel fiber has nine points. This is the ternary-sextic case of [1].

For $(X, L) = (\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(2))$,

$$\sum_{d \geq 0} \binom{2d+3}{3} t^d = \frac{1+6t+t^2}{(1-t)^4}.$$

Thus $(m, c) = (3, 6)$ and a reduced kernel fiber has eight points, recovering the quaternary-quartic case.

Example 9.2 (The square). For $(X, L) = (\mathbb{P}^1 \times \mathbb{P}^1, \mathcal{O}(2, 2))$,

$$\sum_{d \geq 0} (2d+1)^2 t^d = \frac{1+6t+t^2}{(1-t)^3}.$$

Hence $(m, c) = (2, 6)$ and a reduced kernel fiber has eight points. This is the projective toric model associated with the lattice square $[0, 2]^2$.

Example 9.3 (Further Segre examples). The standard Segre embeddings

$$\mathbb{P}^2 \times \mathbb{P}^2 \hookrightarrow \mathbb{P}^8, \quad (\mathbb{P}^1)^3 \hookrightarrow \mathbb{P}^7$$

have Hilbert functions $\binom{d+2}{2}^2$ and $(d+1)^3$, respectively, and both generating functions have numerator $1+4t+t^2$. Thus $c = 4$ and $\deg X = 6 = c + 2$. By contrast, $\mathbb{P}^2 \times \mathbb{P}^3 \hookrightarrow \mathbb{P}^{11}$ has dimension five, codimension six, and degree $\binom{5}{2} = 10$, so it does not satisfy $\deg X = c + 2$.

Remark 9.4 (Scope and novelty). The extreme-rank dichotomy and the minimal SOS length are contained in [2, Proposition 3.5]. The purpose of this note is to give a complete specialized proof under the stronger arithmetically Gorenstein Hilbert-series hypothesis and to package every non-evaluation ray into a finite kernel morphism. The fiberwise reciprocal identities generalize the explicit Cayley–Bacharach formulas of [1, Theorems 6.1 and 7.1] from the two classical Veronese cases to every variety satisfying (1).

10 Conclusion

The Hilbert numerator $1 + ct + t^2$ supplies exactly the algebra needed for the separation theorem. A basepoint-free extreme kernel contains a regular sequence of $m + 1$ linear forms, and the Artinian reduction has Hilbert function $(1, c, 1)$; the perfect Gorenstein pairing forces kernel dimension $m + 1$ and moment rank c . Removing one kernel section produces a one-dimensional arithmetically Gorenstein ring with Hilbert function

$$1, c + 1, c + 2, c + 2, \dots,$$

which controls every finite linear section: its length is $c + 2$, degree-one evaluation has one relation with no zero coefficient, and degree-two evaluation is an isomorphism after quotienting by the section equations. The inertia of the evaluation form permits at most one conjugate pair. The unique relation then determines the radical through the reciprocal identities, and conversely those identities reconstruct a basepoint-free extreme kernel. These facts yield both complete extreme separation of $P_X \setminus \Sigma_X$ and the exact minimal SOS length on the strictly positive boundary.

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